

Nmap Network Scan & Target Reconnaissance Report

Target IP Address: 192.168.1.50

Target Subnet: 192.168.1.0/24

Scanner OS: Kali Linux

1. Phase 1: Host Discovery

To identify live hosts on the target subnet without triggering heavy port scans, a ping sweep is performed.

Command Executed:

```
nmap -sn 192.168.1.0/24
```

Command Explanation:

The `-sn` flag disables port scanning after host discovery, sending ICMP echo requests, TCP SYN/ACK packets, and ARP requests (if local) to determine active IP addresses across the 192.168.1.0/24 network.

2. Phase 2: Service & Version Detection

Once a target IP (192.168.1.50) is discovered, we perform a service version scan to determine what software is listening on open ports.

Command Executed:

```
nmap -sV 192.168.1.50
```

Command Explanation:

The `-sV` flag probes open ports to determine service and version information. It sends specific protocol banners and probes to match responses against Nmap's service detection database.

3. Phase 3: Aggressive Scan & Advanced Reconnaissance

To gather operating system details, traceroute paths, and script detection output, an aggressive scan is performed.

Command Executed:

```
nmap -A 192.168.1.50
```

Understanding Output of nmap -A <target-ip>:

The -A option enables a combination of advanced scanning features in a single flag:

- **OS Detection (-O):** Sends TCP/IP stack fingerprinting packets to deduce the target OS (e.g., Linux 4.X / 5.X, Windows Server 2019).
- **Version Detection (-sV):** Identifies running application protocols and exact software version numbers.
- **Script Scanning (-sC):** Runs the default set of Nmap Scripting Engine (NSE) scripts to check for common vulnerabilities, SSH host keys, HTTP titling, and anonymous FTP access.
- **Traceroute (--traceroute):** Traces the network path to the target host to analyze hop counts and network topology.

4. Open Ports and Services Summary Table

Port	Protocol	State	Service	Version / Details
21	TCP	Open	ftp	vsftpd 3.0.3 (Anonymous login allowed)
22	TCP	Open	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.5
80	TCP	Open	http	Apache httpd 2.4.41 ((Ubuntu))
139	TCP	Open	netbios-ssn	Samba smbd 4.6.2
445	TCP	Open	netbios-ssn	Samba smbd 4.6.2 (Workgroup: WORKGROUP)
3306	TCP	Open	mysql	MySQL 8.0.28-0ubuntu0.2 0.04.3

5. Terminal Execution Screenshot Output

Below is the simulated text screenshot output representing the execution of the full scan on the

lab target environment:

```
root@kali:~# nmap -A 192.168.1.50
```

```
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-06 18:00 PKT
```

```
Nmap scan report for 192.168.1.50
```

```
Host is up (0.00042s latency).
```

```
Not shown: 994 closed tcp ports (reset)
```

```
PORT STATE SERVICE VERSION
```

```
21/tcp open  ftp vsftpd 3.0.3
```

```
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
```

```
|_ -rw-r--r-- 1 0 0 1024 Sep 01 12:00 welcome.txt
```

```
22/tcp open  ssh OpenSSH 8.2p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
```

```
| ssh-hostkey:
```

```
| 3072 48:2c:ea:5e:bf:0b:23:43:0d:2c:94:d9:b3:03:f2:0a (RSA)
```

```
80/tcp open  http Apache httpd 2.4.41 ((Ubuntu))
```

```
|_ http-server-header: Apache/2.4.41 (Ubuntu)
```

```
|_ http-title: Lab Target - Web Portal
```

```
139/tcp open netbios-ssn Samba smbd 4.6.2
```

```
445/tcp open netbios-ssn Samba smbd 4.6.2
```

```
3306/tcp open mysql MySQL 8.0.28-0ubuntu0.20.04.3
```

```
MAC Address: 00:0C:29:8B:2A:1F (VMware)
```

```
Device type: general purpose
```

```
Running: Linux 5.X
```

```
OS CPE: cpe:/o:linux:linux_kernel:5
```

```
OS details: Linux 5.4 - 5.15
```

```
Network Distance: 1 hop
```

```
TRACEROUTE
```

```
HOP RTT ADDRESS
```

```
1 0.42 ms 192.168.1.50
```

```
Nmap done: 1 IP address (1 host up) scanned in 12.48 seconds
```